

MATTHEW L BROWN

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SUMMARY

Full-stack developer with 10+ years of experience designing data-intensive applications, analytics platforms, and cloud infrastructure. Experienced leading projects from business requirements through architecture, development, deployment, and user adoption across healthcare, manufacturing, and e-commerce.

FREELANCE

Software Consultant / Solutions Architect

July 2019 - Present

Selected Client Projects

Population Health Analytics Tool (Highmark Health) [R, SQL, JavaScript]

- Led development of a population health analytics platform used to analyze billions of healthcare records by 1000s of people across the organization
- Developed ELT process to consolidate terabytes of data from 10+ sources (Claims, SDOH, etc.) into **BigQuery**
- Enabled researchers and analysts to perform cohort analysis and real-world evidence studies.
- Enabled the organization to retire 10+ Tableau dashboards and standardize reporting.

Healthcare Platform Cloud Migration [terraform]

- Led migration of multi-service healthcare platform (App Engine, GKE, Cloud Run, Cloud SQL, Pub/Sub) to Azure Container Apps and PostgreSQL Flexible Server.
- Re-architected networking with Azure Front Door, VNet integration, private endpoints, and Managed Identity.
- Identified and remediated critical security risks, including publicly exposed SQL instances.
- Implemented GitHub Actions CI/CD and modular Terraform IaC, enabling repeatable deployments.

Reinsurance Claims Analytics Tool [Python, React, TypeScript]

- Designed and developed an analytics platform for insurance claims, weather, and geospatial data.
- Built Python ETL pipelines to ingest claims data, weather data, and shapefiles into a SQL-based platform.
- Used GeoPandas and Shapely to integrate GIS datasets for catastrophe risk analysis.
- Developed flexible dashboards and optimized SQL queries for real-time access to large datasets

Business Rule Engine for Clinical Interventions (Highmark Health) [R, JavaScript]

- Developed a database-driven system to identify members for clinical interventions
- Designed an intuitive GUI interface so non-technical users could define member cohorts without code.
- Built internal tooling that significantly reduced development time for new intervention cohorts.
- The system served as core functionality for a team of ~50 employees, including ~10 software developers. The core logic that I wrote has not been modified in over six years

Digitizing Iowa Plastics Manufacturer [Microsoft Ecosystem, R Shiny]

- Built a QR-based traceability system linking production, quality, inventory, and shipping data.
- Developed real-time inventory tracking and automated label printing for shop-floor operations.
- Created operational dashboards for production, downtime, quality, and inventory performance.
- Connected machine, quality, and inventory data to reduce scrap and material waste.

CRM Application Suite Optimization for a Health Insurer (Highmark Health) [R, SQL, JavaScript]

- Optimized a suite of R Shiny applications that were slowing down due to rapid growth.
- Centralized duplicate code across the applications into a repository.
- Improve application load times by 10x through optimizing SQL queries.

FOUNDER

Notifi: Back in Stock Alerts

Nov. 2020 - Oct. 2025

*Built and operated a Shopify SaaS application for back-in-stock notifications with **1000+ installs**, **600+ active users**, and **\$1000+ MRR***

Echo - Speech-to-Text Engine

June 2024 - Dec 2024

Built an AI-powered Chrome extension for real-time speech-to-text transcription.

Avisos: PPC Management Copilot

Dec. 2026 - May 2026

Avisos connects your Google Ads account and Shopify store to audit 50+ performance, targeting, tracking, and landing page issues.

EMPLOYMENT

RED OAK STRATEGIC

Washington, D.C.

Senior Data Scientist

April 2017- June 2019

Developed analytics software for financial services startup

- Led a team of three in developing 30+ custom analytical applications using **R Shiny**
- Led development of portions of an online platform, from collecting product requirements and data modeling to building the production applications using **.NET** and **SQL Server**
- Developed predictive models on large datasets using machine learning techniques such as gradient boosting, feature engineering and ensemble methods to inform client decision-making

BOOZ ALLEN HAMILTON

Washington, D.C.

Senior Consultant

June 2015- April 2017

Management consultant focused on implementing large-scale technology projects for Federal clients

- Developed strategies to maximize adoption of **agile software development** best practices as a **project manager** on a congressionally mandated multi-million dollar project
- **Scrum master** for an **agile software development** project to migrate [DOL.gov](https://www.dol.gov) to Drupal
- Received 'High Five Award' for work on an analytical dashboard built in **Tableau** for a Federal client

JOHN DEERE

Waterloo, IA

Geometric Coordinator

Nov. 2011-Aug. 2013

Quality analyst for design engineering division

TECHNICAL SKILLS

Languages: Python, R, SQL, TypeScript, JavaScript, C#

Frameworks: Django, React, FastAPI, R Shiny, .NET, React

Databases: PostgreSQL, BigQuery, SQL Server, MongoDB, dbt

Cloud & DevOps: Google Cloud, Azure, AWS, Terraform, Docker, GitHub Actions

Coursework / Certifications: *Introduction to R for Data Science, The Analytics Edge, Explore Data Science, Advanced Analytics, Statistical Methods of Process Improvement*

EDUCATION

HENRY B. TIPPIE SCHOOL OF MANAGEMENT, UNIVERSITY OF IOWA

Iowa City, IA

Master of Business Administration, Supply Chain and Analytics

May 2015

UNIVERSITY OF IOWA

Iowa City, IA

BSE Biomedical Engineering

May 2011